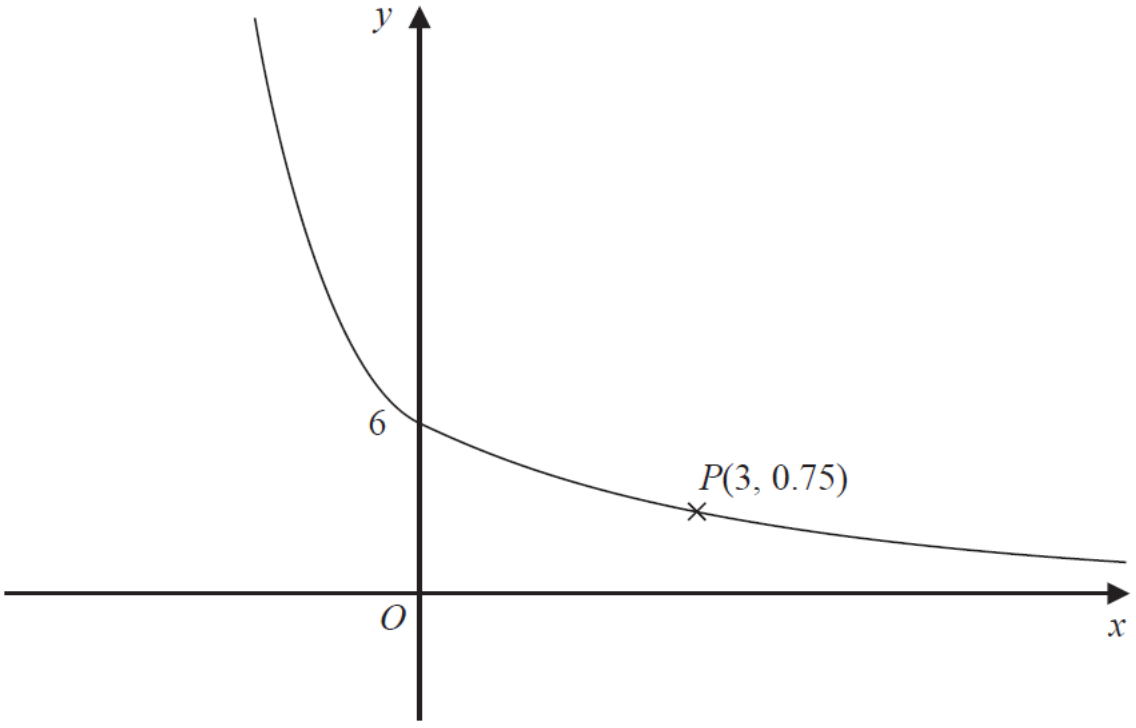
21 The diagram shows a sketch of the graph with equation $y = ab^{-x}$



P is a point on the graph with coordinates (3, 0.75)

Work out the value of a and the value of b.

At $x = 0$, $y = 6$: $6 = ab^0 = a$, so $a = 6$.

1 mark
Use y-intercept.

At P: $0.75 = 6b^{-3} = 6 \div b^3$

1 mark
Substitute P.

$b^3 = 6 \div 0.75 = 8$

Since $2^3 = 8$, $b = 2$.

1 mark
a = 6, b = 2.

$a =$ .....

$b =$ .....

(Total for Question 21 is 3 marks)